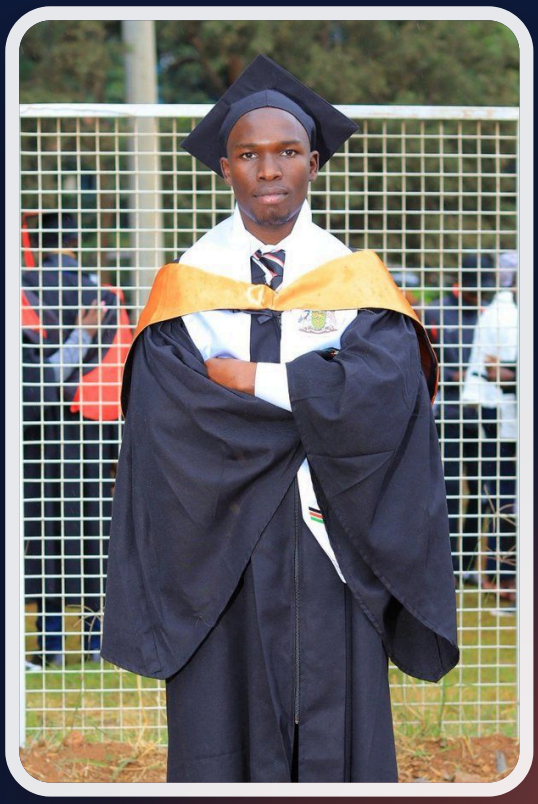
EO

Nexus CAD Studio

ENGINEERING · GRAPHIC DESIGN · TRAINING

Portfolio 2026

Engineering precision. *Creative* design.



Edison Ochieng, Mechanical Design Engineer & Graphic Designer

Founder of Nexus CAD Studio · Nairobi, Kenya

WHATSAPP
+254 708 639 473

WEBSITE
edisonengineer.github.io/MY-PAST-PROJECTS

LINKEDIN
in/edison-ochieng

ABOUT ME

Hello, I'm *Edison*.

Mechanical Engineering graduate from the University of Nairobi with hands-on CAD experience in SolidWorks, Autodesk Inventor and Fusion 360. I produce 3D models, assemblies and 2D technical drawings for industrial prototypes, and run Nexus CAD Studio, a CAD and graphic design studio that has trained over 100 people. My final-year project used coupled FE simulation of sheet metal forming, reflecting strong manufacturing and design fundamentals. I am eager to contribute to product development, prototyping and technical support.

100+
people trained in CAD

3+ yrs
hands-on CAD experience

Honorable Mention
Autodesk Designathon 2025

WHAT I DO

- CAD & engineering design: SolidWorks, Inventor, Fusion 360, AutoCAD, Revit
- Manufacturing processes & DFM principles
- Simulation & FEA: SolidWorks Simulation, Abaqus
- Technical drawing & documentation
- Prototype fabrication & lab testing
- Graphic design: Illustrator, Photoshop, InDesign, Premiere Pro
- CAD training & facilitation
- Leadership, collaboration & communication

EXPERIENCE

Founder
Nexus CAD Studio, Nairobi · Dec 2024 – Present

Industrial Attachment, Mechanical Engineering
Kenya Industrial Research and Development Institute (KIRDI), Nairobi · Jun – Aug 2025 (12 weeks)

Autodesk Fusion Design
KnowledgePoint Limited · Jun – Jul 2025

Leader, Workshop & Exhibition Planning
Women in Engineering Students Summit 4.0 (WIESS 4.0), University of Nairobi · 2025 – 2026

Academic Secretary
University of Nairobi Students Association (UNSA) · Apr – Nov 2024

EDUCATION

BSc Mechanical Engineering
University of Nairobi · 2021 – 2026

KCSE, Mean Grade A
Aoch Muga Secondary School, Homa Bay · 2017 – 2020

CERTIFICATIONS

Certified SOLIDWORKS Associate (CSWA) Cert Prep · LinkedIn Learning

CSWA Exam Preparation (SOLIDWORKS) · Udemy

Autodesk Inventor Design Certification · LinkedIn Learning

Autodesk Designathon 2025, Honorable Mention · Autodesk

Autodesk Designathon 2026, Participant · Autodesk

SOLIDWORKS 2024 Essential Training · LinkedIn Learning

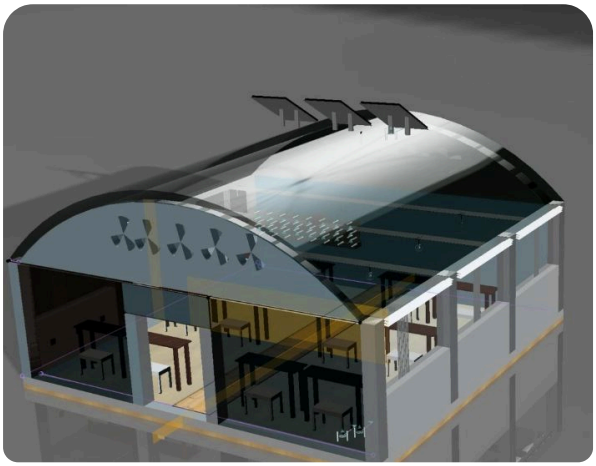
Premiere Pro Essential Training · LinkedIn Learning

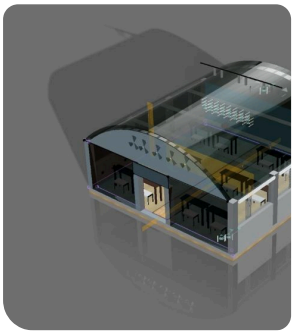
LANGUAGES

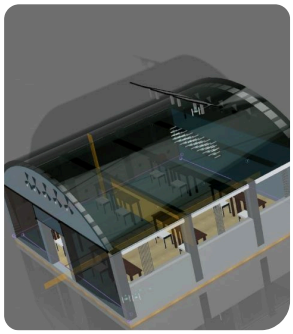
English (full professional) · French (native / bilingual) · Kiswahili (fluent)

SELECTED WORK

Projects







ENGINEERING · 2025

Sustainable Outdoor Hub

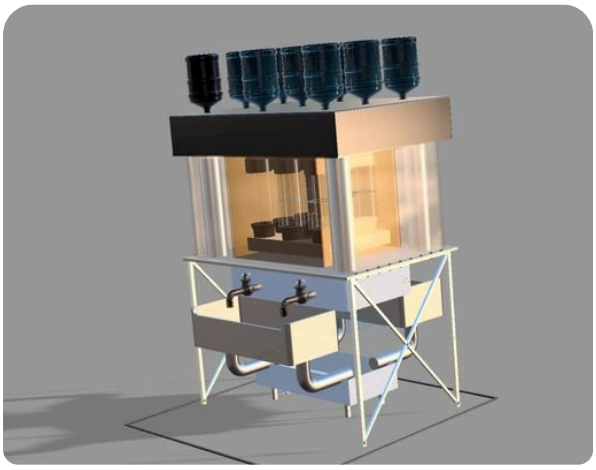
Autodesk Designathon 2025 · Honorable Mention

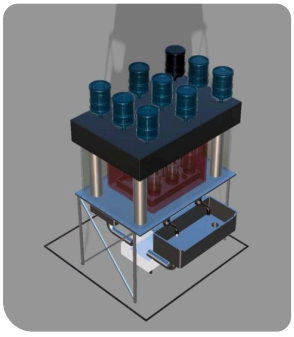
A team design for an accessible, all-season outdoor workspace on a university campus, powered by renewable energy. The entry earned an Honorable Mention at the Autodesk Designathon 2025.

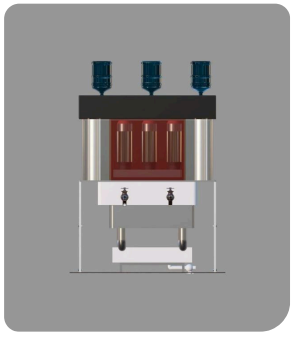
- 9.5 m × 9.5 m structure supported on 3 m concrete pillars
- 6 m radius recycled-polycarbonate roof with integrated solar panels
- Ergonomic charging stations for students working outdoors
- Validated through 3D modelling and structural simulation in Fusion 360

Fusion 360 · Sustainable design · Structural simulation

edisonengineer.github.io/MY-PAST-PROJECTS/projects/designathon-hub.html







ENGINEERING · 2026

Emergency Hygiene Station

Autodesk Designathon 2026

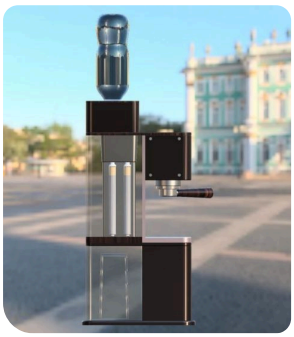
A compact, rapidly deployable hygiene station for communities displaced by natural disasters, designed for the Autodesk Designathon 2026 brief: a place to wash hands, get filtered water and keep up basic hygiene, all within a 2 m × 2 m footprint.

- Header tank fed by nine inverted 19-litre carboys, the water containers most easily available after a disaster
- Enclosed filtration chamber with three filter columns, visible through clear acrylic panels
- Twin stainless-steel basins with taps on opposite sides, so two people can wash their hands at once
- U-bend drains carry used water to a separate wastewater tank with a drain valve below

Fusion 360 · Humanitarian design · Water filtration

edisonengineer.github.io/MY-PAST-PROJECTS/projects/designathon-2026-hygiene-station.html







ENGINEERING · 2026

Espresso Coffee Machine

Product design · May – June 2026

A compact countertop espresso machine designed from concept to photorealistic render: the full product, from the water supply and internal layout to the brew group and drip tray, modelled as one assembly in Fusion 360.

- Twin inverted-bottle water reservoirs that feed the machine from the top
- Clear acrylic side panel that shows the twin filter cartridges and internal layout
- Group head with a wooden-handled portafilter over a perforated stainless drip tray
- Material palette of dark bamboo trim, anodised aluminium, stainless steel and ABS

Fusion 360 · Product design · Rendering

edisonengineer.github.io/MY-PAST-PROJECTS/projects/coffee-machine.html







ENGINEERING · 2025

Sheet Metal Fabrication at KIRDI

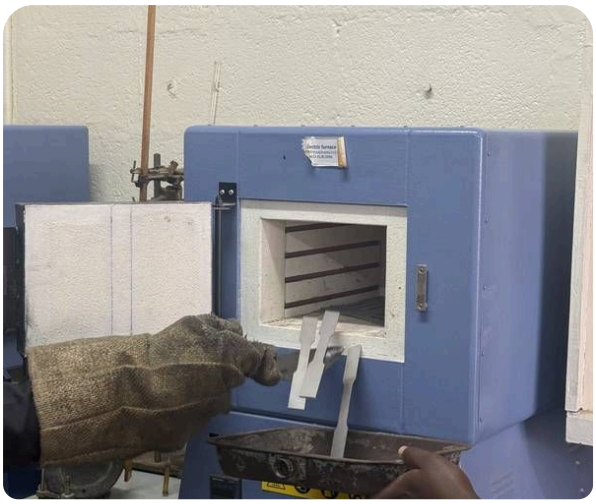
Industrial attachment · Jun – Aug 2025

A 12-week attachment at the Kenya Industrial Research and Development Institute's Engineering Research and ICT Department, designing and fabricating real components for projects that support Kenyan MSMEs.

- Designed and fabricated jiko stove components from sheet metal, applying DFM principles and manufacturing tolerances
- Helped design and build a poultry de-feathering machine, from concept to component iteration and assembly
- 3D models, assemblies and 2D technical drawings in SolidWorks, Inventor and Fusion 360
- CNC laser cutting and bending of L-beams, car bumpers and machine casings to precise tolerances

DFM · Sheet metal · CNC laser cutting · SolidWorks

edisonengineer.github.io/MY-PAST-PROJECTS/projects/kirdi-fabrication.html







ENGINEERING · 2026

Forming Limit Prediction for AM AlSi10Mg

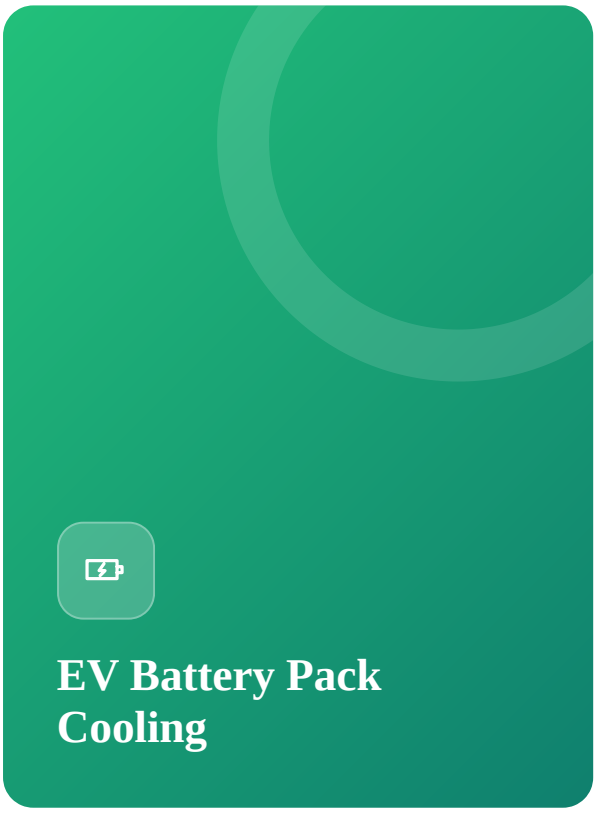
Final-year project · 2025 – 2026

Predicting the forming limit curve of additively manufactured AlSi10Mg sheet through a coupled GTN–Hill'48 simulation of the Nakajima test, so that sheet-metal parts can be checked for failure before they are made.

- Coupled damage mechanics (GTN) with Hill'48 anisotropic plasticity
- Heat-treated and tensile-tested AlSi10Mg specimens in the lab to calibrate the GTN–Hill'48 model
- Simulated the Nakajima stretch-forming test to build the forming limit curve
- Practical failure prediction for sheet-metal parts before manufacture

FEA · Damage mechanics · Sheet metal forming · Tensile testing

edisonengineer.github.io/MY-PAST-PROJECTS/projects/final-year-project.html



ENGINEERING

EV Battery Pack Cooling

Freelance client project · Design & thermal simulation

Designed and simulated a complete electric-vehicle battery cooling pack for a private client, covering 3D modelling, assembly and thermal performance analysis.

- Full 3D model and assembly of the battery cooling pack
- Thermal simulation to keep the battery cells within their required operating temperature range
- Iterated the cooling geometry and layout based on the results
- Delivered a complete design package: 3D models, assemblies and simulation results

SolidWorks · Inventor · Thermal simulation

edisonengineer.github.io/MY-PAST-PROJECTS/projects/ev-battery-cooling.html



GRAPHIC DESIGN · 2026

Camp Meeting Appreciation

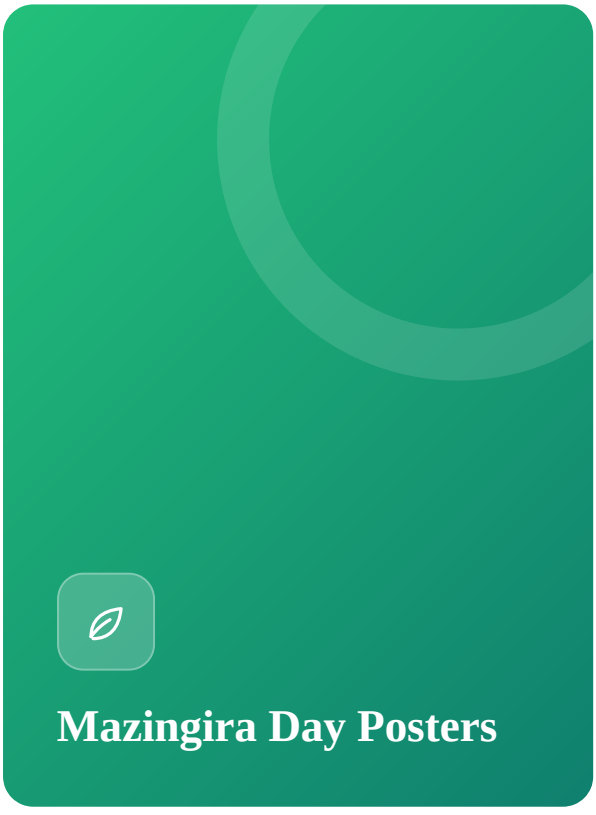
Certificate series · SDA Church Ngong Road

A series of appreciation certificates recognising ministries for their contributions to the church's Camp Meeting fundraising goal.

- Consistent layout across the whole series, personalised per recipient
- Bold typography with a warm, celebratory colour palette

Illustrator · Certificates · Print

edisonengineer.github.io/MY-PAST-PROJECTS/projects/camp-meeting.html



GRAPHIC DESIGN

Mazingira Day Posters

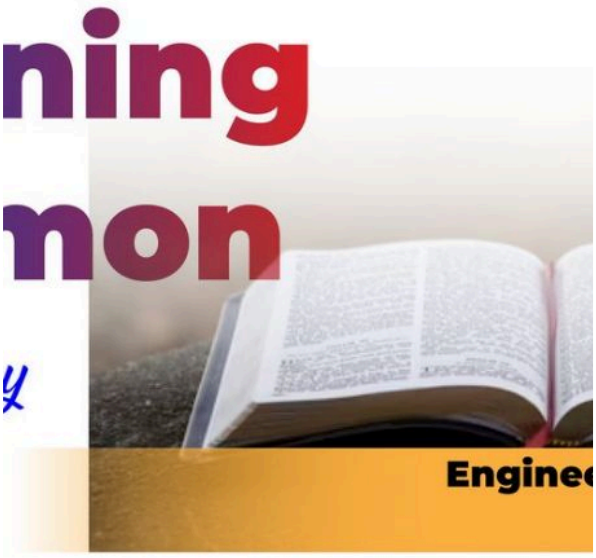
Environmental awareness campaign

Campaign posters for Mazingira Day, encouraging care for the environment with clear messaging and vibrant visuals.

- Poster series designed for print and social media

Poster design · Campaign · Illustrator

edisonengineer.github.io/MY-PAST-PROJECTS/projects/mazingira-day.html



GRAPHIC DESIGN

Evening Sermon

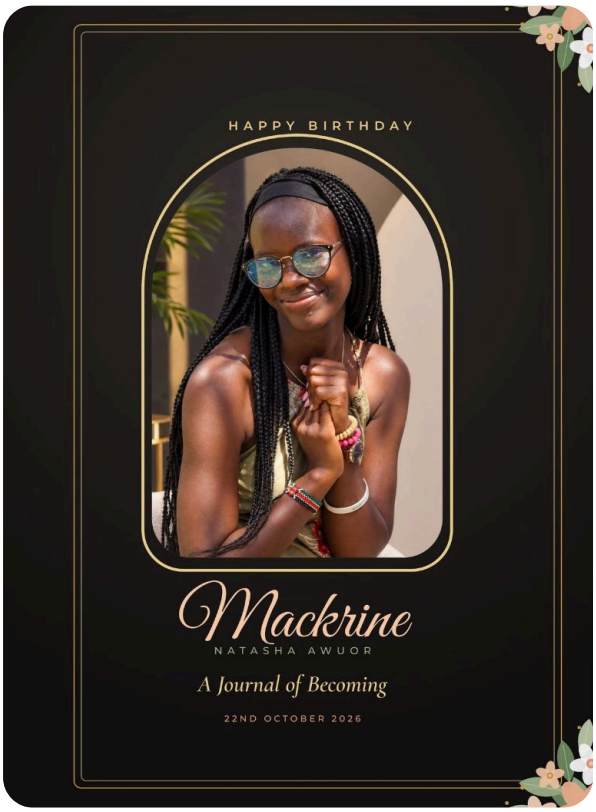
Event graphic

A bold, widescreen event graphic announcing a daily evening sermon series.

- Gradient headline typography paired with photography

Illustrator · Social media · Event

edisonengineer.github.io/MY-PAST-PROJECTS/projects/evening-sermon.html



GRAPHIC DESIGN · 2026

A Journal of Becoming (Mackrine Natasha Awuor Birthday Journal)

Client Project 2026

A 120-page personalised birthday journal, designed in InDesign for a long-standing client's 22nd October 2026 birthday. It guides her through reflection, dreams and future planning in a refined white-and-gold style.

- 120-page print-ready journal in Adobe InDesign
- Three sections: About Me, Dreams & Visions, and The Woman Called Mackrine
- Custom layouts for vision boards, travel plans and a dream-career page
- Cohesive Great Vibes, Cormorant Garamond and Montserrat typography

Indesign · Illustrator

edisonengineer.github.io/MY-PAST-PROJECTS/projects/a-journal-of-becoming-mackrine-natasha-awuor-birthday-journal.html

SERVICES & PRICING

How I can *help*.

3D Part Model
A single part modelled from your sketch, dimensions or physical sample.

From KES 1,000

Assembly & Drawings
Multi-part assemblies with manufacturing-ready drawings.

From KES 4,500

Simulation Study
FEA or thermal analysis to validate a design before it is built.

From KES 4,500

Social Media Pack
A set of on-brand posts and stories for your campaign or event.

From KES 3,500

Poster & Print
Posters, flyers, certificates and banners that stand out.

From KES 500

Brand Identity
A logo and visual identity that makes your business recognisable.

From KES 7,500

CAD Training
One-on-one or group training, online or in person in Nairobi.

From KES 4,500

Let's work together.

Send your project brief on the website or WhatsApp me. I reply within a day.

WHATSAPP
+254 708 639 473

START A PROJECT
edisonengineer.github.io/MY-PAST-PROJECTS/#brief

M-PESA BUY GOODS TILL
8438334